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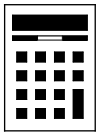
Functional Skills Level 2
MATHEMATICS

Paper 2 Calculator

Monday 13 January 2020 Afternoon Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments.



Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
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For Examiner's Use	
Question	Mark
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Full Marks / Exemplary (100% Pass · Model Solutions)

· Assessment: January 2020 - Paper 2

· Subject: AQA Functional Skills Maths Level 2

· Complete accredited step-by-step models and tutor guidance

· Exam Board: AQA (Ofqual Regulated Reform Standard)

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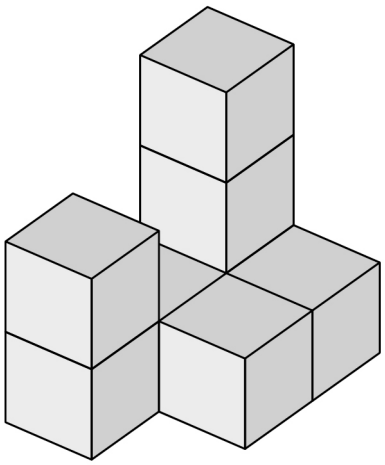
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Section A

Answer **all** questions in the spaces provided.

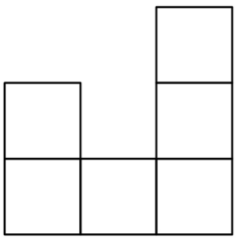
1 The diagram shows a 3-D shape.



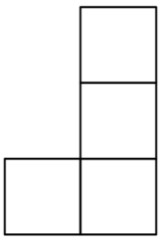
Which is the plan view of the shape?
Circle the correct letter.

[1 mark]

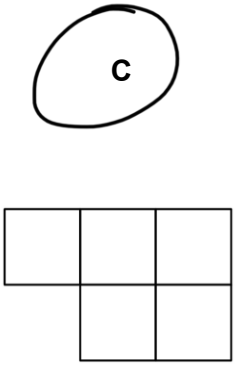
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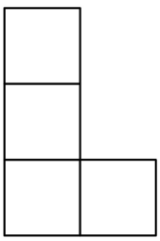
B



C



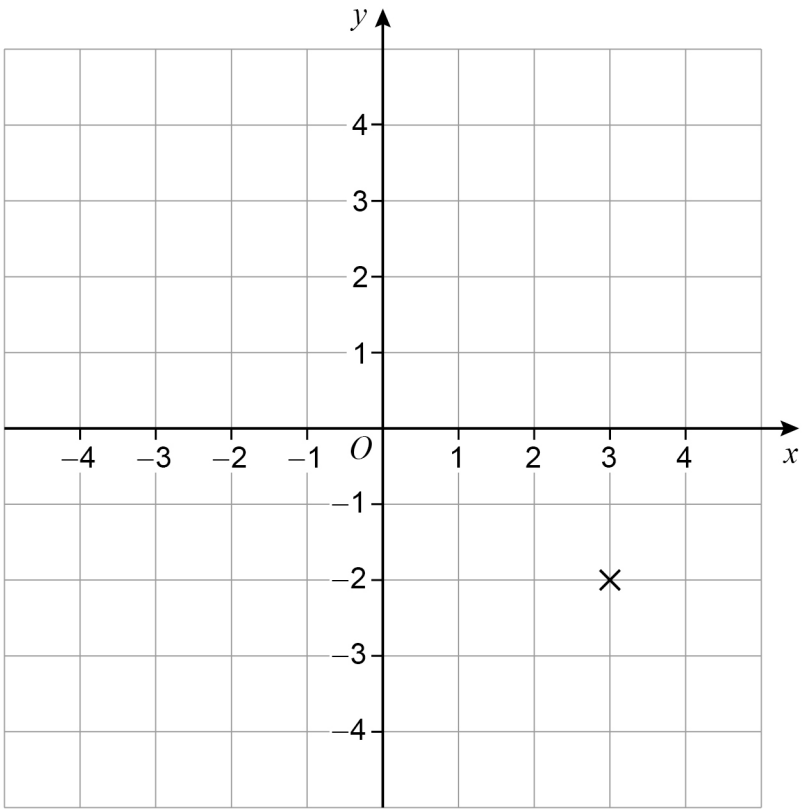
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2 Write down the coordinates of the point marked ×.

[1 mark]



(x, y)

Answer (3 , −2)

3 Increase 1670 by 27%

[2 marks]

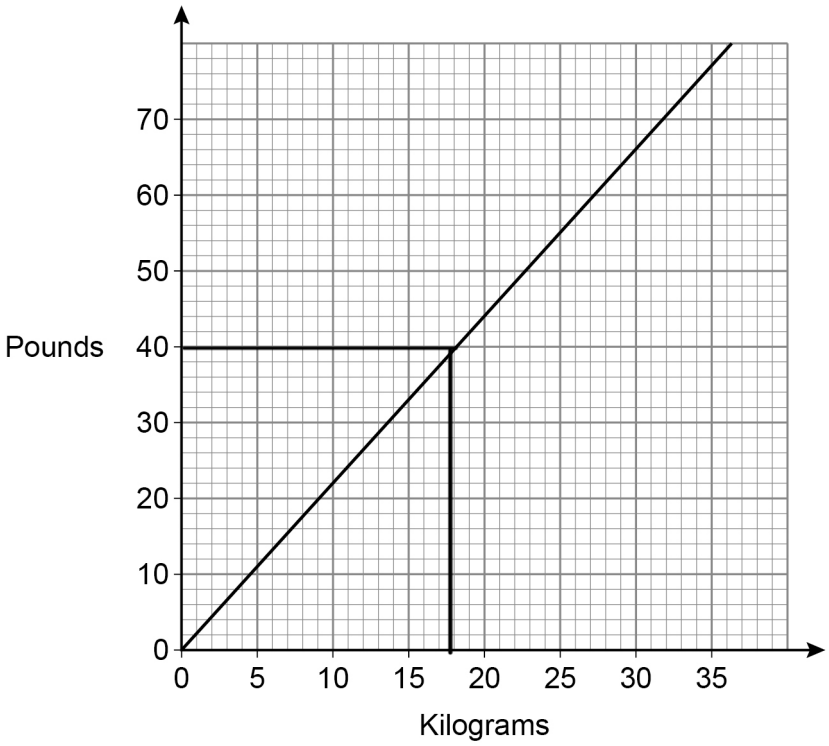
1670 × 1.27 = 2120.9

Answer 2120.9

Turn over ►

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4 The graph can be used to convert between pounds and kilograms.



Use the graph to convert 40 pounds to kilograms.

[2 marks]

Answer 18 kilograms

Do not write outside the box

5 Calculate $1\frac{3}{4} + 2\frac{4}{5}$

[1 mark]

$\frac{7}{3} + \frac{14}{5} = 4\,{}^{11}/_{20}$

Answer $4\,{}^{11}/_{20}$

6 Complete the table to show equivalent fractions, decimals and percentages.

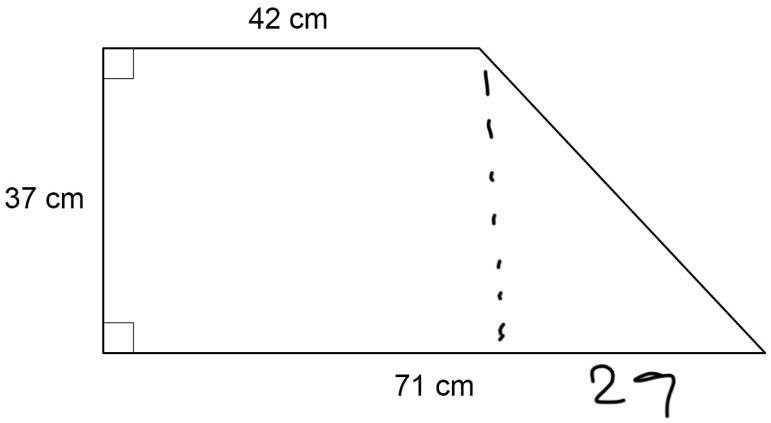
[2 marks]

Fraction	Decimal	Percentage
$\frac{3}{8}$	0.375	37.5%
$\frac{15}{100}$	0.15	15%

Turn over for the next question

Do not write outside the box

7 A shape is made from a rectangle and a triangle.



Not drawn accurately

Work out the area of the shape.

[3 marks]

$71 - 42 = 29$

$42 \times 37 = 1554$

$\frac{1}{2} \times 29 \times 37 = 536.5$

$1554 + 536.5 = 2090.5$

Answer 2090.5 cm^2

12

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Section B

Answer **all** questions in the spaces provided.

8 Fashion blogger

Alex is a fashion blogger.

Companies pay her to promote their goods.

8 (a) Alex writes 30 posts promoting Company A.

The table shows the number of views of each post.

Number of views, x	Frequency	mid point	mid x Fq
$0 \leqslant x < 30\,000$	5	15,000	75,000
$30\,000 \leqslant x < 60\,000$	9	45,000	405,000
$60\,000 \leqslant x < 90\,000$	12	75,000	900,000
$90\,000 \leqslant x < 120\,000$	4	105,000	420,000
	Total = 30		1,800,000

Alex also writes 23 posts promoting Company B.

The total number of views of these posts is 1 223 600

Which company receives the **higher** average number of views, A or B?

You **must** show your working.

[4 marks]

1223,600 ÷ 23 = 53,200

1,800,000 ÷ 30 = 60,000

Company A

Turn over ►

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Some of the posts were about clothing and the rest were about shoes.

The table shows the number of each.

	Company A	Company B
Clothing	21	13
Shoes	9	10

total
34
19
53

One of the posts is chosen at random.

8 (b) Work out the probability that it is about clothing.

[1 mark]

Answer 34/53

8 (c) Work out the probability that it is from Company A **and** is about shoes.

[1 mark]

Answer 9/53

Do not write outside the box

8 (d) In a year, Alex wrote 1460 posts promoting companies.
She was paid £75 for each post that had at least 65 000 views.
In total, she was paid £36 000

What percentage of her posts had at least 65 000 views?

[3 marks]

$36,000 \div 75 = 480$

$\frac{480}{1460} \times 100 = 32.9$

Answer 32.9 %

9

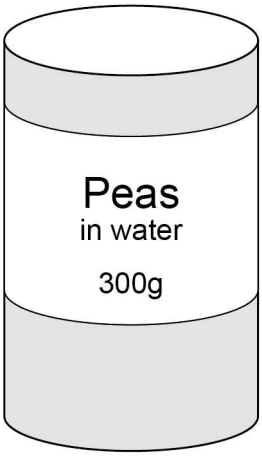
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9 **Tinned food**

A company makes tinned food.

9 (a) A tin is filled with peas and water.



The contents of the tin have a total mass of 300 grams.

mass of peas : mass of water = 3 : 2

240 of these tins are packed into a box.

Work out the total mass of peas in the box.
Give your answer in kilograms.

[5 marks]

3+2 =5 300÷5=60

60x3 =180g of peas per tin

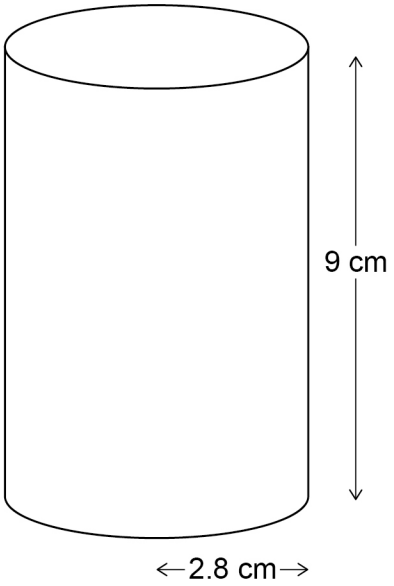
180g =0.18kg

0.18 x240 =43.2

Answer 43.2 kilograms

Do not write outside the box

9 (b) A tin for carrots is a cylinder with radius 2.8 cm and height 9 cm



When filled, one sixth of the volume of the tin of carrots is water.

The company uses 425 litres of water each day in tins of carrots.

1 litre = 1000 cm^3

How many tins of carrots does the company make each day?

[5 marks]

$2.8^2 \times \pi = 24.6\,cm^2$

$24.6 \times 9 = 221.7\,cm^3$

$\frac{1}{6} \times 221.7 = 36.9\,cm^3$ water

$425\,L = 425{,}000\,cm^3$

$425{,}000 \div 36.9 = 11504$

Answer 11504

10

Turn over ►

Do not write outside the box

10 **University student**

Marco is about to start university.

The table shows his total expected income for the **year**.

	Amount (£)
Student loan	8700
Part-time job	4800

Marco budgets for the following expenses each **month** for 12 months.

	Amount per month (£)
Accommodation	589.00
Living expenses	186.00
Entertainment	65.00
Travel	87.50

10 (a) Marco will save any income that he does not spend.

How much does he expect to save in the **year**?

[4 marks]

8700 + 4800 = 13,500

589 + 186 + 65 + 87.50

= 927.50

927.50 × 12 = 11130

13500 − 11130 = 2370

Answer £ 2370

Do not write outside the box

10 (b) On Marco's 14th birthday, his parents put £2000 into a bank account for him.
The account pays compound interest at a rate of 1.5% per annum.
Marco can access the account on his 21st birthday.

How much will be in the account on his 21st birthday?

[3 marks]

$21 - 14 = 7$

$1.5\% = 0.015$

$1.015^{7} \times 2000 = 2219.69$

Answer £ 2219.69

7

Turn over for the next question

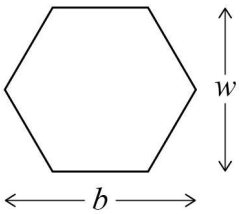
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11 Jewellery

11 (a) Mo makes and sells jewellery.

He makes pendants in the shape of hexagons.

The hexagons are made of glass and have wire around the perimeter.



Not drawn accurately

Here is a formula for the perimeter, P, of the hexagon.

$$P = 6 \times \sqrt{\left(\frac{b}{4}\right)^2 + \left(\frac{w}{2}\right)^2}$$

b is the length of the pendant

w is the width of the pendant

Mo makes pendants with length 6.4 cm and width 5.5 cm

He buys the wire in reels with 4 metres of wire on each reel.

How many pendants can Mo make using one reel of wire?
You **must** show your working.

[5 marks]

$\left(\frac{6.4}{4}\right)^2 + \left(\frac{5.5}{2}\right)^2 = 1.6^2 + 2.75^2$

$= 10.1225$

$\sqrt{10.1225} = 3.18$

$6 \times 3.18 = 19.09$

$4m = 400cm$

$400 \div 19.09 = 20.95$

Answer 20

Do not write outside the box

11 (b) Mo makes some pendants.

The glass in each pendant is red, blue or yellow.

$\frac{2}{11}$ of the pendants are red

blue pendants and yellow pendants are in the ratio 2 : 1

What fraction of the pendants are blue?

[4 marks]

$1 - \frac{2}{11} = \frac{9}{11}$

$2 + 1 = 3$ $\quad \frac{9}{11} \div 3 = \frac{3}{11}$ yellow

$\frac{3}{11} \times 2 = \frac{6}{11}$ blue

Answer $\frac{6}{11}$

9

Turn over for the next question

Do not write outside the box

12 **Football match**

Steph supports Leicester City Football Club.

She and three friends are going to watch the team play at West Ham United.

12 (a) The table shows information about the three parts of their journey.

Drive 108 miles	Average speed 48 miles per hour
Train	Maximum 34 minutes
Walk to stadium	Maximum 25 minutes

What is the latest time they can leave to be sure of getting to the stadium by 2.30 pm?
You **must** show your working.

[4 marks]

$S = d/t \quad t = d/s$

Drive:

108/48 = 2.25 hours

= 135 mins

135 + 34 + 25 = 194 mins

= 3 hours 14 mins

2:30 pm - 3 hours 14 mins

= 11:16 am

Answer 11:16 am

Do not write outside the box

12 (b) Steph works out the cost of her trip, including the return journey.

She will use her car to drive 108 miles **each way**.

She and her 3 friends will share the cost of fuel equally.

Her car travels 12.5 miles per litre of fuel.

Fuel costs 128.8p per litre.

Steph will also pay

£3 for a return train ticket

£30 for a match ticket

£8 for food.

Work out the total amount Steph will pay for her trip.

[6 marks]

108 x 2 = 216

216 ÷ 12.5 = 17.28 litres

17.28 x 128.8 = 2225.664 p

2225.664 ÷ 4 people = 556.416p

556.416p ≈ £5.56

5.56 + 3 + 30 + 8 = 46.56

Answer £ 46.56

Question 12 continues on the next page

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12 (c) Steph looks at all the match results Leicester City have had at West Ham United.

Leicester win	West Ham win	Draw
12	32	15

Steph says,

"Based on these results, the probability of a Leicester win is more than 20%"

Is she correct?
You **must** show your working.

[3 marks]

$12 + 32 + 15 = 59$

$12/59 \times 100 = 20.3\%$

Yes, $20.3\% > 20\%$

13

END OF QUESTIONS

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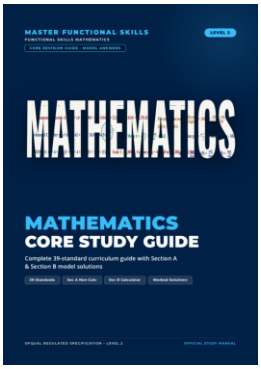
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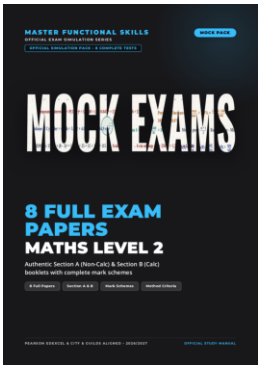


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